

CompTIA

A+ Core 2

Exam 220-1202

A+ 220-1202

Module 21



Using Data Security

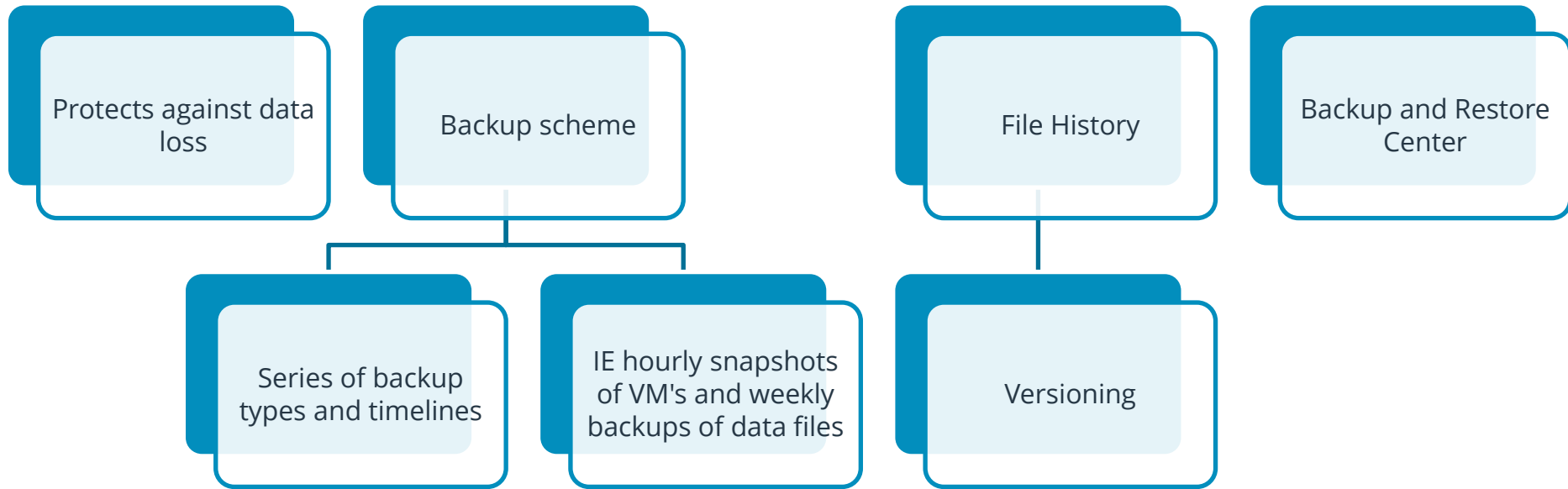
Prepare for A+ Core 2 by:

- Implementing backup and recovery
- Explaining data handling best practices
- Explaining the basics of Artificial Intelligence

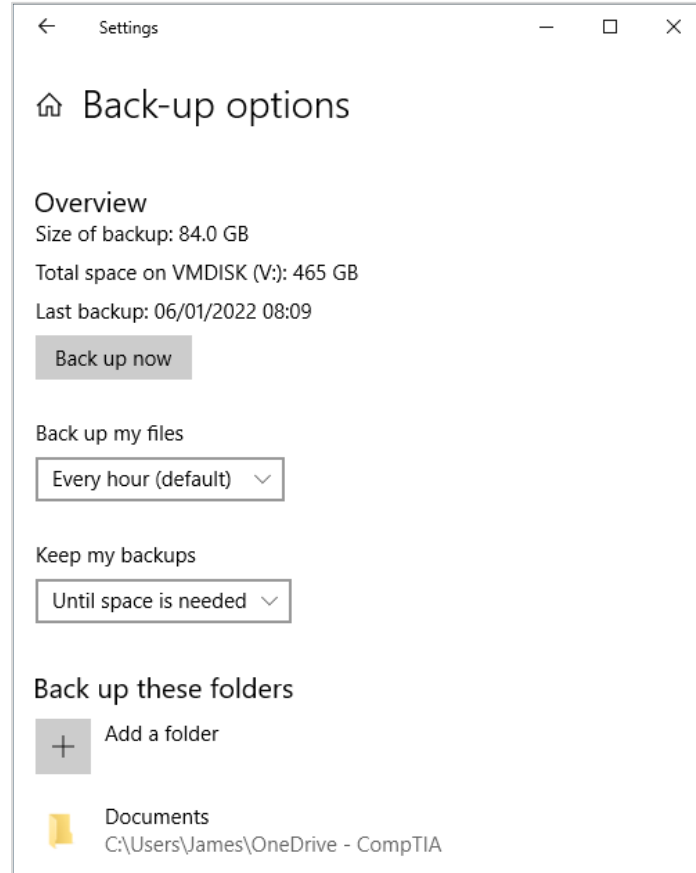
Lesson 21.1

Data Backup and Recovery

Backup Operations (Slide 1 of 2)



Backup Operations (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Backup Methods (Slide 1 of 2)

- Frequency- how often
- Retention- how long to hold on to backup file
- Backup chains
 - Full- all data
 - Incremental- only changes since last backup of any type
 - Differential- changes since last FULL backup
- Synthetic full backup
 - Series of backups that append changes from incremental backup to the end of the original full backup
 - IE: Full backup on the first of month at 7am and then incremental backups every 3 hours for remainder of month

Backup Methods (Slide 2 of 2)

Type	Data Selection	Backup Job Time and Storage Requirement	Recovery Complexity	Archive Attribute
Full	All selected data regardless of when it was previously backed up	High	Low (single job)	Cleared
Incremental	New files and files modified since last backup job	Low	High (multiple jobs)	Cleared
Differential	New files and files modified since last full backup job	Moderate	Moderate (two jobs)	Not cleared

Backup Media Requirements

Media reuse

- Magnetic tape, internal/external hard drives, NAS/SAN, cloud

Grandfather-Father-Son (GFS)

On site vs off site storage

3-2-1 rule

- Popular scheme consists of
- 3 full copies of data(includes production system), 2 different media types, offline and offsite stored backup
- IE: Production system, External hard drive backup stored on site, and a backup of data to the cloud

Backup Testing and Recovery Best Practices

- Always test backups will provide restoration of data in the event of a loss
- Verify data after backup
- Test periodically to ensure procedural compliance
- Recovery options
 - In-place
 - Alternative system/site location

Lab Activity

- Lab: Back Up the Computer
(available in CertMaster Learn and CertMaster Perform)
- Lab: Restore Data from File History
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Support Backup and Restore Operations
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Data Backup and Recovery

- Backup Operations
- Backup Methods
- Backup Media Requirements
- Backup Testing and Recovery Best Practices

Lesson 21.2

Data Handling Best Practices

Regulated Data Classification

- PII
 - US Privacy Act and GDPR regulatory compliance
- PGI
 - Social security, passport, driver's license
- PHI
 - Protected health information
 - HIPAA compliance in US
- Credit card and bank card transactions
 - PCI-DSS
- Best practices and requirements

Prohibited Content and Licensing Issues

- Information not applicable to work
 - Obscene or illegally copied/pirated
- End-user license agreements (EULA)
- License compliance
 - Personal vs corporate use
 - Single user vs multiple user license
- Open-source license
 - Does not necessarily indicate free cost to use
- Digital rights management
- Non-disclosure agreement (NDA)

Incident Response

- Standard procedures and policy to follow when something occurs like a data breach or system outage
- Incident response plan
- CIRT/CERT/CSERT
 - Team of responders identifies specifically to handle various situations

Data Integrity and Preservation

- Forensics
 - Collection of evidence from a computer system or network to a standard that will be accepted in a court of law
 - Certified technicians and facilities for processing
- Document, document, document
 - Pictures, videos, notes
- Order of volatility
- Chain of custody
 - Positive, documented, control over evidence to prevent tampering and documents collection, storage, processing, and even transport of said evidence

Data Destruction Methods

Sanitization

Re-formatting may not be sufficient

Erasing/wiping

Low level format

- Secure erase
- Instant secure erase

Disposal and Recycling Outsourcing Concepts

Drilling

Shredding

Incineration

Degaussing

- Only works on magnetic media like HDD and magnetic tape, not SSD or optical media

Certificate of destruction/recycling

Lesson Review: Data Handling Best Practices

- Regulated Data Classification
- Prohibited Content and Licensing Issues
- Incident Response
- Data Integrity and Preservation
- Data Destruction Methods
- Disposal and Recycling Outsourcing Concepts

Lesson 21.3

Artificial Intelligence

Application Integration



AI can analyze data and provide recommendations



Use of third-party providers like website add-on chatbot



NLP, ML, Computer Vision

Policy

Organizational policy should be designed to dictate the legal and ethical use or banning of AI solution use

Acceptable use policy

May restrict what corporate data can and cannot be shared with authorized AI integrations

Limitations

Bias- garbage in, garbage out

Hallucinations- wrong info generated without regard for facts

Accuracy- trust, but verify

Private vs. Public

Private- for the sole use within the organization
for business purposes



Public- available for anyone to utilize

CHATGPT, Gemini, and Bard are examples

Lab Activity

- Challenge Live Lab: Resolve Incident Response Tickets
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Artificial Intelligence

- Application Integration
- Policy
- Limitations
- Private vs. Public